

IV. METHOD

A. Method Overview

Chapter III formulates the considered problem as a joint user association and beam power control problem under user association, beam power, satellite power budget, and EPFD constraints. The association variables are discrete, while the beam power variables are continuous, making the problem a mixed-integer nonlinear optimization problem. In addition, user association affects beam loading, downlink SINR, time-division resource sharing, and satisfaction, while beam power affects both user service quality and the aggregate EPFD at the GS. Directly solving the global optimum for every time slot is therefore computationally demanding.

To reduce online decision complexity, we propose Safe-Beam Assisted Power-Release Relay (SAPR-R). SAPR-R uses the predictable LEO satellite motion, GS location, GSO direction, and beam footprints to precompute beam-level relation records for all time slots in the observation window. For each time slot, the record includes the critical satellite, the north and south helper satellites, the closed beams on the critical satellite, the safe beams that remain active with default power, and the beam-pair mappings between critical beams and helper beams. This precomputation depends only on satellite geometry, beam footprints, and beam-level EPFD calculation, rather than user distribution, and can therefore be completed before online user reassociation.

During online operation, SAPR-R loads the beam relation record of the current time slot and makes user-aware reassociation decisions based on user locations, current associations, and service satisfaction. First, Safe-Beam Assisted Helper Relief (SAHR) lets critical safe beams take over selected users originally served by helper release beams. A reassociation is accepted only when it does not reduce the satisfaction of the transferred user or the users already served by the critical safe beam. The transmit power no longer required by the helper release beams is then added to the available power budget of the helper satellite.

Next, the helper satellite allocates its available power budget to helper relay beams according to relay demand, so that these beams can provide additional service capacity for users affected by closed beams. Finally, Beam-Pair Local Relay (BPLR) reassociates closed-beam users to feasible helper relay beams within the precomputed beam-pair mappings. Candidate users are served sequentially according to relay link quality and remaining relay capacity.

In summary, SAPR-R consists of the following steps:

- **Time-slot beam relation record:** Precompute the critical/helper satellites, closed/safe beams, and beam-pair mappings for each time slot.
- **Safe-Beam Assisted Helper Relief (SAHR):** Use critical safe beams to take over selected helper users without reducing satisfaction, thereby releasing helper satellite power.
- **Helper relay beam boosting:** Allocate the available helper satellite power budget to helper relay beams according to relay demand.
- **Beam-Pair Local Relay (BPLR):** Reassociate closed-beam users to feasible helper relay beams within each beam pair

according to relay link quality and remaining relay capacity.

B. Time-Slot Beam Relation Record

For each time slot t , SAPR-R constructs a beam relation record $\mathcal{R}_B(t)$, which stores the beam-level information used by the online reassociation and relay procedures. Figure 1 illustrates the time-varying beam status of the critical satellite and its helper satellites. The record is defined as

$$\mathcal{R}_B(t) = (s^C(t), s^N(t), s^S(t), \mathcal{B}_c^{\text{off}}(t), \mathcal{B}_c^{\text{safe}}(t), \mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t), \mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)).$$

In this record, $s^C(t)$, $s^N(t)$, and $s^S(t)$ denote the critical satellite, north helper satellite, and south helper satellite, respectively. $\mathcal{B}_c^{\text{off}}(t)$ denotes the set of closed beams on the critical satellite, and $\mathcal{B}_c^{\text{safe}}(t)$ denotes the set of safe beams that remain active with default power. $\mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t)$ and $\mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)$ denote the safe-beam-to-helper-release-beam mapping and the closed-beam-to-helper-relay-beam mapping, respectively.

1) Critical and Helper Satellite Identification: In each time slot t , the critical satellite $s^C(t)$ is identified as the visible LEO satellite that contributes the largest EPFD risk at the GS. Since the EPFD violation is mainly caused by the critical satellite, beam shutdown is applied only to beams on $s^C(t)$, while helper and other visible satellites remain available for service continuity. The along-track north and south neighboring satellites of $s^C(t)$ are selected as helper satellites, denoted by $s^N(t)$ and $s^S(t)$, respectively.

2) EPFD-Driven Beam Status Determination: For each beam b on the critical satellite, the beam-level EPFD risk score is defined as

$$q_{c,b}(t) = \text{EPFD}_{s^C,b}(t).$$

The critical beams are sorted in descending order of $q_{c,b}(t)$. A greedy shutdown rule is then applied: at each step, the beam with the highest EPFD risk score is shut down until the aggregate EPFD at the GS satisfies the applicable EPFD limit.

The beams shut down by this procedure form the closed-beam set $\mathcal{B}_c^{\text{off}}(t)$, and the remaining active beams form the safe-beam set:

$$\mathcal{B}_c^{\text{safe}}(t) = \mathcal{B}_{s^C} \setminus \mathcal{B}_c^{\text{off}}(t).$$

Because this beam status determination depends only on beam-level EPFD contributions and the EPFD limit, it is independent of user distribution and can be precomputed before online user reassociation.

3) Beam-Pair Mapping Construction: Given the closed-beam set $\mathcal{B}_c^{\text{off}}(t)$ and the safe-beam set $\mathcal{B}_c^{\text{safe}}(t)$, SAPR-R constructs beam-pair mappings based on footprint overlap between the critical satellite and its helper satellites.

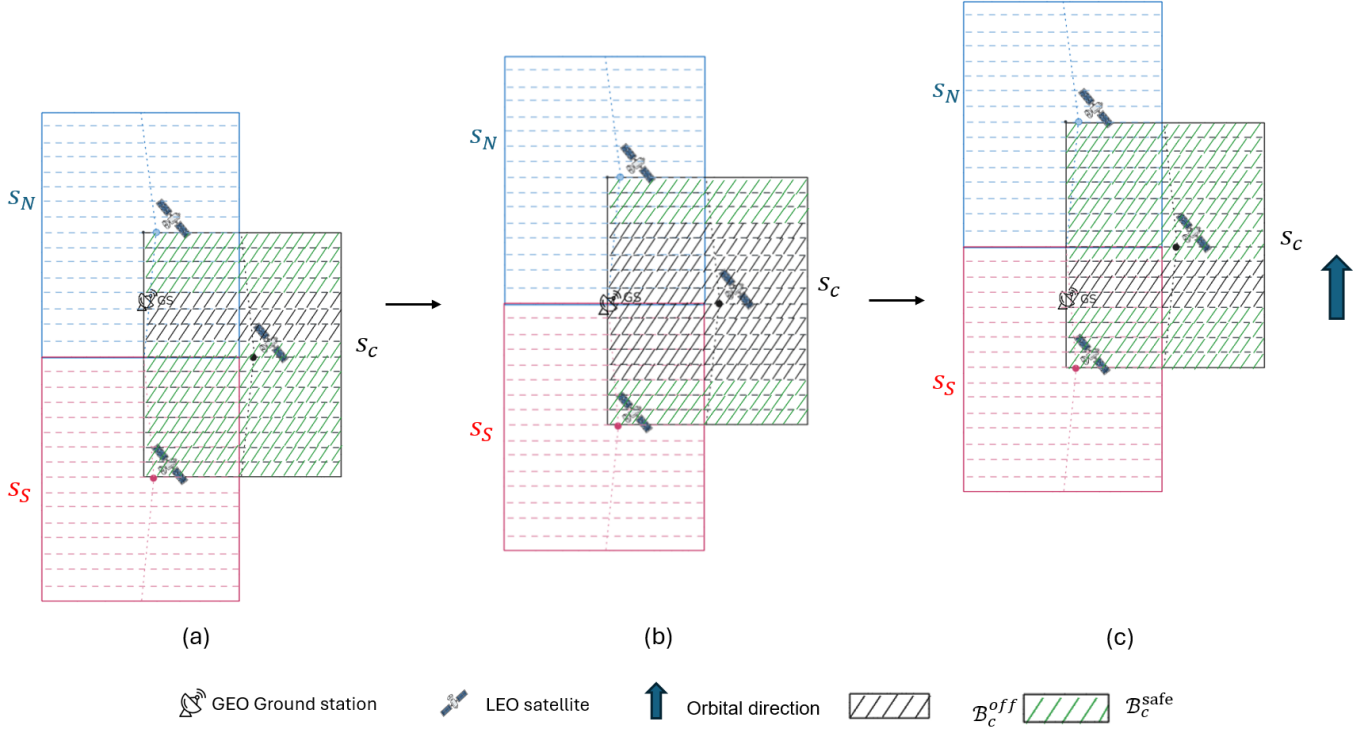


Fig. 1. Three time slots as the along-track LEO satellites (s^N , s^c , and s^S) pass the protected GSO ground station. In region S_C on the critical satellite, black-hatched beams are $\mathcal{B}_c^{\text{off}}(t)$ (closed beams) and green-hatched beams are $\mathcal{B}_c^{\text{safe}}(t)$ (safe beams). Panels (a)–(c) show earlier to later time slots.

For SAHR, the safe-beam-to-helper-release-beam mapping is defined as

$$\begin{aligned} \mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t) = \{ & (b_c^{\text{safe}}, s_h, b_h^{\text{rel}}) : \\ & b_c^{\text{safe}} \in \mathcal{B}_c^{\text{safe}}(t), \\ & s_h \in \{s^N(t), s^S(t)\}, \\ & b_h^{\text{rel}} \in \mathcal{B}_{s_h}, \\ & \mathcal{F}_{s^c, b_c^{\text{safe}}}(t) \cap \mathcal{F}_{s_h, b_h^{\text{rel}}}(t) \neq \emptyset \}. \end{aligned}$$

This mapping identifies the helper beams whose users can be reassociated to critical safe beams.

For BPLR, the closed-beam-to-helper-relay-beam mapping is defined as

$$\begin{aligned} \mathcal{M}_{\text{off} \rightarrow \text{relay}}(t) = \{ & (b_c^{\text{off}}, s_h, b_h^{\text{relay}}) : \\ & b_c^{\text{off}} \in \mathcal{B}_c^{\text{off}}(t), \\ & s_h \in \{s^N(t), s^S(t)\}, \\ & b_h^{\text{relay}} \in \mathcal{B}_{s_h}, \\ & \mathcal{F}_{s^c, b_c^{\text{off}}}(t) \cap \mathcal{F}_{s_h, b_h^{\text{relay}}}(t) \neq \emptyset \}. \end{aligned}$$

This mapping identifies the helper beams that can serve users originally located under critical closed beams.

Figure 2 illustrates the two mappings stored in $\mathcal{R}_B(t)$: $\mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t)$ for SAHR and $\mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)$ for BPLR. These mappings rely only on footprint overlap and therefore can be applied to general multi-beam LEO systems, as long as the beam footprints and helper-satellite geometry are known. In the considered along-track helper scenario, the overlap is local and can be handled within predefined beam pairs, avoiding a global user-to-beam matching process.

For each helper satellite, helper beams are assigned to one of two roles in a time slot: release beams for SAHR and relay beams for BPLR. To avoid role conflict, the two sets are kept disjoint:

$$\mathcal{B}_{s_h}^{\text{rel}}(t) \cap \mathcal{B}_{s_h}^{\text{relay}}(t) = \emptyset, \quad \forall s_h \in \{s^N(t), s^S(t)\}.$$

C. Safe-Beam Assisted Helper Relief

After loading $\mathcal{R}_B(t)$, the online procedure starts with SAHR, as illustrated in Fig. 3. SAHR does not serve critical closed-beam users directly. Instead, it uses critical safe beams to take over selected users originally served by helper release beams, so that the corresponding helper satellite can make more power available for subsequent relay-beam boosting.

For each mapping tuple $(b_c^{\text{safe}}, s_h, b_h^{\text{rel}}) \in \mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t)$, SAPR-R identifies the helper users that can be transferred from the helper release beam to the critical safe beam. Let u_h denote a helper user. The candidate helper-user set is defined as

$$\begin{aligned} \mathcal{V}_{\text{rel}}(b_c^{\text{safe}}, s_h, b_h^{\text{rel}}, t) = \\ \{ & u_h \in \mathcal{U}(t) : x_{u_h, s_h, b_h^{\text{rel}}}(t) = 1, \\ & u_h \in \mathcal{F}_{s^c, b_c^{\text{safe}}}(t) \}. \end{aligned}$$

The first condition means that user u_h is currently served by helper release beam b_h^{rel} on helper satellite s_h . The second condition means that the same user is also covered by the critical safe beam b_c^{safe} . Therefore, users in $\mathcal{V}_{\text{rel}}(b_c^{\text{safe}}, s_h, b_h^{\text{rel}}, t)$ are eligible to be reassociated from the helper release beam to the critical safe beam.

Beam-pair mapping visualization for a given time slot t

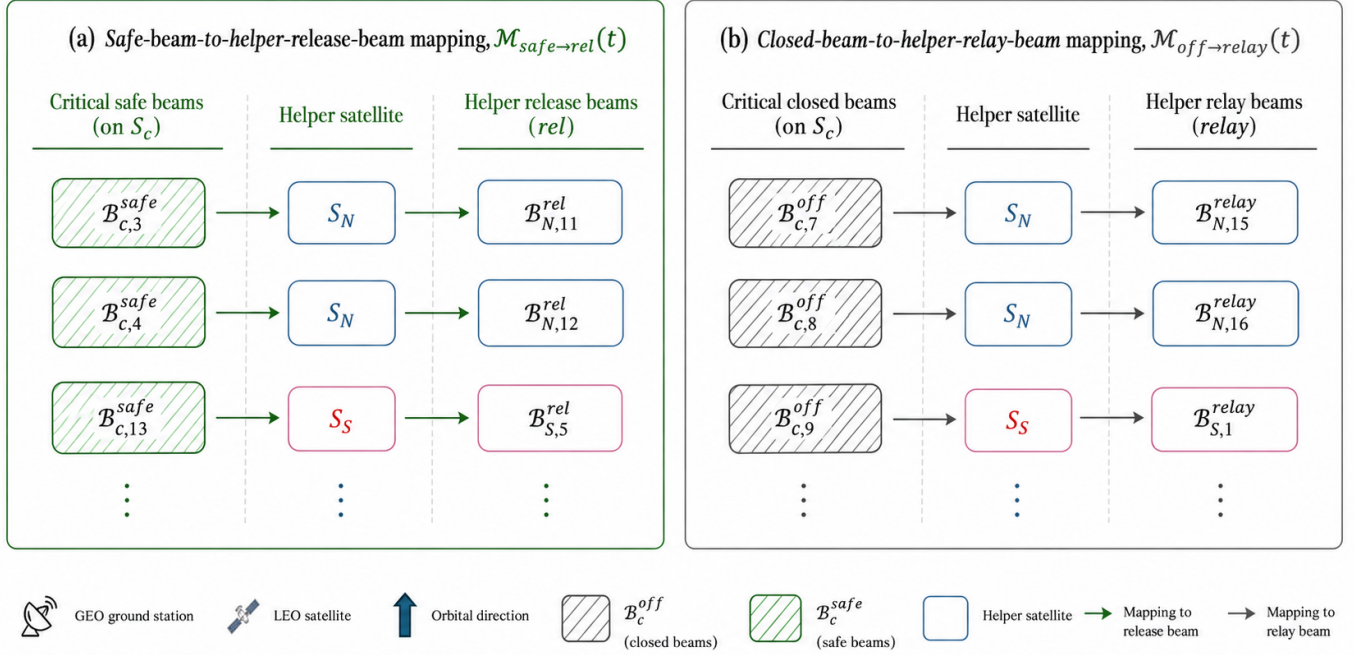


Fig. 2. Beam-pair mapping tables at time slot t . (a) $\mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t)$: critical safe beams on S^C mapped to helper release beams on S^N/S^S (SAHR). (b) $\mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)$: critical closed beams mapped to helper relay beams (BPLR).

If candidate helper user u_h is reassocated to b_c^{safe} , the reduction in the required transmit power of the helper release beam is defined as

$$\Delta P_{u_h}^{\text{rel}}(t) = P_{b_h^{\text{rel}}}^{\text{req,old}}(t) - P_{b_h^{\text{rel}}}^{\text{req,new}}(u_h, t).$$

Here, $P_{b_h^{\text{rel}}}^{\text{req,old}}(t)$ denotes the required transmit power of helper release beam b_h^{rel} before transferring u_h , while $P_{b_h^{\text{rel}}}^{\text{req,new}}(u_h, t)$ denotes the required transmit power of the same beam after u_h is removed from that beam. Therefore, $\Delta P_{u_h}^{\text{rel}}(t)$ represents the amount of helper transmit power released by transferring u_h to the critical safe beam.

Candidates are ranked in descending order of $\Delta P_{u_h}^{\text{rel}}(t)$, so users that release more helper power are considered first. A trial reassociation is accepted only when the transferred helper user and the original users on the critical safe beam do not lose satisfaction:

$$\begin{aligned} \text{sat}_{u_h}^{\text{new}}(t) &\geq \text{sat}_{u_h}^{\text{old}}(t), \\ \text{sat}_{u_s}^{\text{new}}(t) &\geq \text{sat}_{u_s}^{\text{old}}(t), \quad \forall u_s \in \mathcal{U}_{b_c^{\text{safe}}}^{\text{ori}}(t). \end{aligned}$$

Accepted transfers form the set $\mathcal{A}_{\text{rel}}(s_h, t)$. The total released power on helper satellite s_h is

$$P_{s_h}^{\text{rel}}(t) = \sum_{u_h \in \mathcal{A}_{\text{rel}}(s_h, t)} \Delta P_{u_h}^{\text{rel}}(t).$$

The available power budget for relay-beam boosting consists of the unused satellite power under the default beam-power profile and the power released by SAHR:

$$P_{s_h}^{\text{pool}}(t) = P_{s_h}^{\text{unused}}(t) + P_{s_h}^{\text{rel}}(t),$$

where

$$P_{s_h}^{\text{unused}}(t) = P_{s_h}^{\text{max}} - \sum_{b \in \mathcal{B}_{s_h}} P_{s_h, b}^{\text{def}}.$$

Helper beams that are not selected as release beams or relay beams keep their default transmit powers.

D. Helper Relay Beam Boosting

After SAHR, each helper satellite s_h uses its available relay power budget $P_{s_h}^{\text{pool}}(t)$, defined in the previous subsection, to increase the transmit power of helper relay beams. For helper satellite s_h , the relay beam set is defined as

$$\begin{aligned} \mathcal{B}_{s_h}^{\text{relay}}(t) &= \{b_h^{\text{relay}} : \exists b_c^{\text{off}} \in \mathcal{B}_c^{\text{off}}(t) : \\ &\quad (b_c^{\text{off}}, s_h, b_h^{\text{relay}}) \in \mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)\}. \end{aligned}$$

This set contains the helper beams on s_h that are mapped to at least one closed beam on the critical satellite and can therefore be used for BPLR.

To estimate the relay demand of each helper relay beam, SAPR-R first identifies the candidate closed-beam users that can be served by that beam. Let u_c denote a user originally served by a closed beam on the critical satellite. For helper relay beam b_h^{relay} , the candidate user set is defined as

$$\begin{aligned} \mathcal{U}_{\text{cand}}(b_h^{\text{relay}}, t) &= \\ &\{u_c \in \mathcal{U}(t) : \exists b_c^{\text{off}} \in \mathcal{B}_c^{\text{off}}(t) : \\ &\quad (b_c^{\text{off}}, s_h, b_h^{\text{relay}}) \in \mathcal{M}_{\text{off} \rightarrow \text{relay}}(t), \\ &\quad x_{u_c, S^C, b_c^{\text{off}}}(t) = 1, u_c \in \mathcal{F}_{s_h, b_h^{\text{relay}}}(t)\}. \end{aligned}$$

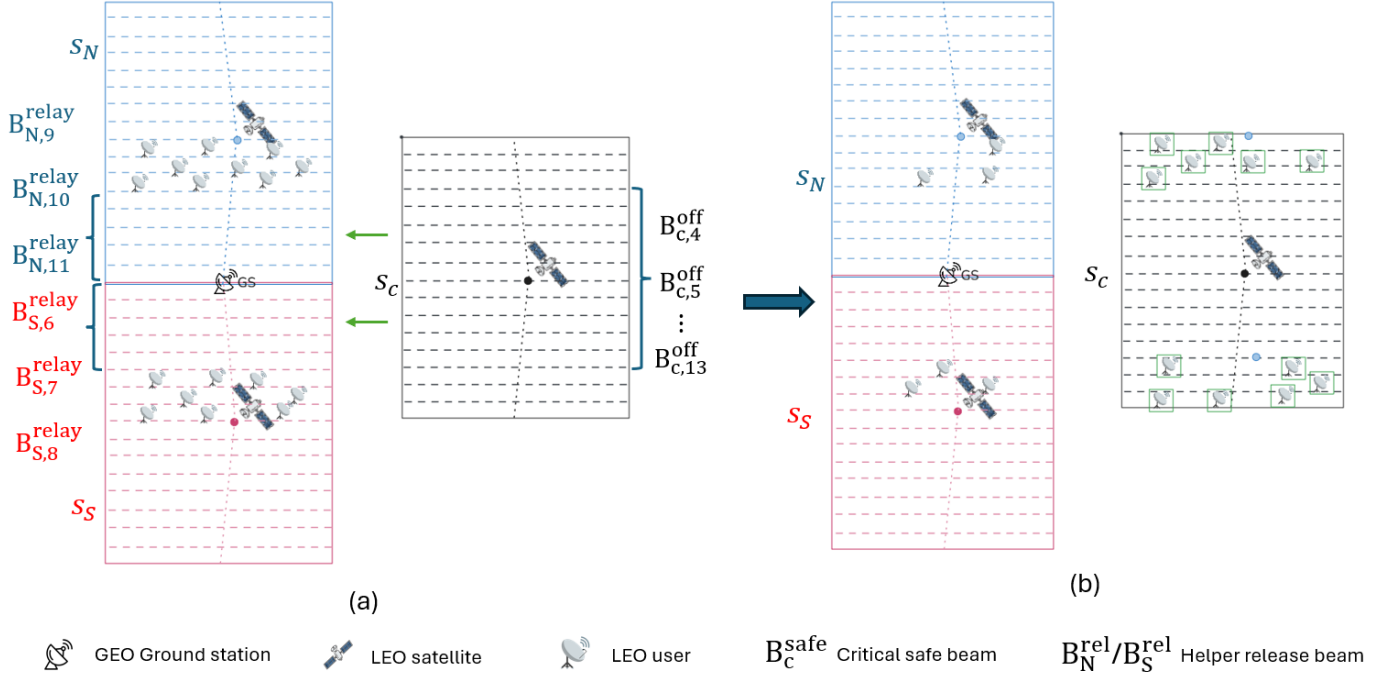


Fig. 3. Illustration of Safe-Beam Assisted Helper Relief (SAHR). Critical safe beams take over eligible helper users from helper release beams, and the released power is returned to the helper satellite power pool.

The condition $x_{u_c, s_c, b_c^{\text{off}}}(t) = 1$ means that user u_c is originally associated with the closed beam b_c^{off} on the critical satellite. The footprint condition means that the same user is also covered by helper relay beam b_h^{relay} . Therefore, $\mathcal{U}_{\text{cand}}(b_h^{\text{relay}}, t)$ represents the closed-beam users that are eligible to be relayed by b_h^{relay} .

The relay demand of helper relay beam b_h^{relay} is defined as the number of its candidate users:

$$D_{b_h^{\text{relay}}}^{\text{relay}}(t) = |\mathcal{U}_{\text{cand}}(b_h^{\text{relay}}, t)|.$$

This demand value is used only to prioritize relay-beam boosting. The actual user reassociation is performed later in BPLR after the final transmit power of each helper relay beam is determined.

For a helper relay beam b_h^{relay} , let $\Delta P_{b_h^{\text{relay}}}^{\text{boost, max}}(t)$ denote the additional power required to increase its transmit power from the default value to the beam-level maximum:

$$\Delta P_{b_h^{\text{relay}}}^{\text{boost, max}}(t) = P_{s_h, b_h^{\text{relay}}}^{\text{max}} - P_{s_h, b_h^{\text{relay}}}^{\text{def}}.$$

If $P_{s_h}^{\text{pool}}(t)$ is sufficient to raise all helper relay beams to their beam-level maximum powers, all relay beams are fully boosted. Otherwise, helper relay beams are sorted in descending order of $D_{b_h^{\text{relay}}}^{\text{relay}}(t)$, and beams with higher relay demand receive additional power first.

For each helper relay beam, if the remaining available relay power budget is sufficient, its transmit power is increased to $P_{s_h, b_h^{\text{relay}}}^{\text{max}}$. If the remaining budget is positive but insufficient for full boosting, the remaining power is assigned as a partial boost above $P_{s_h, b_h^{\text{relay}}}^{\text{def}}$. This avoids leaving unused relay power and increases the available capacity for BPLR.

After this step, each helper relay beam proceeds to BPLR with its final transmit power.

E. Beam-Pair Local Relay

After relay-beam boosting, BPLR performs the actual re-association of closed-beam users to helper relay beams, as illustrated in Fig. 4. For each mapping tuple $(b_c^{\text{off}}, s_h, b_h^{\text{relay}}) \in \mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)$, BPLR considers the users in $\mathcal{U}_{\text{cand}}(b_h^{\text{relay}}, t)$ that were originally served by the closed beam b_c^{off} . Since $\mathcal{U}_{\text{cand}}(b_h^{\text{relay}}, t)$ already includes the coverage condition of the helper relay beam, these users are eligible to be reassociated to b_h^{relay} .

Let u_c denote a closed-beam user considered in this beam pair. For each candidate user u_c , let $R_{u_c, b_h^{\text{relay}}}(t)$ denote the achievable service rate if u_c is served by helper relay beam b_h^{relay} using the final transmit power obtained after relay-beam boosting. Candidates are ranked in descending order of $R_{u_c, b_h^{\text{relay}}}(t)$, so users with better relay link quality are considered first.

Let $C_{b_h^{\text{relay}}}^{\text{rem}}(t)$ denote the remaining relay capacity of helper relay beam b_h^{relay} after preserving the service of its original helper users. This remaining capacity is allocated sequentially to the sorted closed-beam users. If $C_{b_h^{\text{relay}}}^{\text{rem}}(t)$ can satisfy the demand $D_{u_c}(t)$ of user u_c , the user is reassociated to b_h^{relay} and its satisfaction is updated. If the remaining capacity is positive but insufficient to fully satisfy the next user, partial capacity is assigned to that user and the beam-pair process stops. When the remaining capacity is exhausted, no further users are reassociated in this beam pair.

All relay decisions follow the single-association constraint. A relayed user is moved from the critical closed beam to the

helper relay beam and is served by only one beam in the time slot.

F. Overall Algorithm

The online procedure uses a single time-slot-based algorithm. In each time slot, SAPR-R first runs SAHR and relay-beam boosting, then runs BPLR using the updated helper relay-beam powers. The complete procedure is given in Algorithm 1.

Algorithm 1 SAPR-R Online Operation

Input: $\{\mathcal{R}_B(t)\}_{t=1}^T$, initial $\mathbf{x}(t)$, $\mathcal{U}(t)$, $D_u(t)$

Input: $P_{s,b}^{\text{def}}, P_{s,b}^{\text{max}}, P_{s,b}^{\text{max}}, \mathcal{F}_{s,b}(t)$, capacity/satisfaction model

Output: $\mathbf{x}^{\text{final}}(t)$, $P_{s,b}^{\text{final}}(t)$, $\text{sat}_u^{\text{final}}(t)$

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1: for  $t = 1$  to  $T$  do
2:   Load  $\mathcal{R}_B(t)$ ; initialize beam powers per default profile and closed/safe beam status
3:   Initialize  $P_{s,b}^{\text{rel}}(t) = 0$  for each  $s_b \in \{s^N(t), s^S(t)\}$ 
4:   Stage 1: SAHR
5:   for all  $(b_c^{\text{safe}}, s_h, b_h^{\text{rel}}) \in \mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t)$  do
6:     Build  $\mathcal{V}_{\text{rel}}(b_c^{\text{safe}}, s_h, b_h^{\text{rel}}, t)$  using  $\mathbf{x}(t)$  and beam footprints
7:     Compute  $\Delta P_{u_h}^{\text{rel}}(t)$  for each  $u_h \in \mathcal{V}_{\text{rel}}(b_c^{\text{safe}}, s_h, b_h^{\text{rel}}, t)$ 
8:     Sort candidate helper users by  $\Delta P_{u_h}^{\text{rel}}(t)$  descending
9:     for all  $u_h$  in sorted order do
10:      Trial reassociation of  $u_h$  from  $b_h^{\text{rel}}$  to  $b_c^{\text{safe}}$ 
11:      if transferred helper user and original users on  $b_c^{\text{safe}}$  do not lose satisfaction then
12:        Accept the reassociation
13:        Update  $\mathbf{x}(t)$ , satisfaction values, and  $P_{s,b}^{\text{rel}}(t)$ 
14:      end if
15:    end for
16:  end for
17:  Stage 1 continued: helper relay beam boosting
18:  for all  $s_b \in \{s^N(t), s^S(t)\}$  do
19:    Compute unused power  $P_{s_b}^{\text{unused}}(t)$  under default beam-power profile
20:    Compute relay power budget  $P_{s_b}^{\text{pool}}(t) = P_{s_b}^{\text{unused}}(t) + P_{s_b}^{\text{rel}}(t)$ 
21:    Extract helper relay beam set  $\mathcal{B}_{s_b}^{\text{relay}}(t)$  from  $\mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)$ 
22:    for all  $b_h^{\text{relay}} \in \mathcal{B}_{s_b}^{\text{relay}}(t)$  do
23:      Build  $\mathcal{U}_{\text{cand}}(b_h^{\text{relay}}, t)$ 
24:      Compute relay demand  $D_{b_h^{\text{relay}}}^{\text{relay}}(t) = |\mathcal{U}_{\text{cand}}(b_h^{\text{relay}}, t)|$ 
25:      Compute  $\Delta P_{b_h^{\text{relay}}}^{\text{boost,max}}(t)$  for boosting  $b_h^{\text{relay}}$  to beam-level maximum
26:    end for
27:    Sort helper relay beams by  $D_{b_h^{\text{relay}}}^{\text{relay}}(t)$  descending
28:    for all  $b_h^{\text{relay}}$  in sorted order do
29:      Allocate  $P_{s_b}^{\text{pool}}(t)$  to  $b_h^{\text{relay}}$ : full boost, partial boost, or default power
30:      Update final transmit power of  $b_h^{\text{relay}}$ 
31:    end for
32:  end for
33:  Stage 2: Beam-Pair Local Relay
34:  for all  $(b_c^{\text{off}}, s_h, b_h^{\text{relay}}) \in \mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)$  do
35:    Select from  $\mathcal{U}_{\text{cand}}(b_h^{\text{relay}}, t)$  users originally served by  $b_c^{\text{off}}$ 
36:    Compute remaining relay capacity of  $b_h^{\text{relay}}$  after preserving original helper users
37:    Compute  $R_{u_c, b_h^{\text{relay}}}(t)$  for each candidate  $u_c$  using final power of  $b_h^{\text{relay}}$ 
38:    Sort candidate closed-beam users by  $R_{u_c, b_h^{\text{relay}}}(t)$  descending
39:    for all  $u_c$  in sorted order do
40:      if remaining relay capacity can satisfy  $D_{u_c}(t)$  then
41:        Reassociate  $u_c$  to  $b_h^{\text{relay}}$ 
42:        Update  $\text{sat}_{u_c}(t)$  and remaining relay capacity
43:      else if remaining relay capacity  $> 0$  then
44:        Assign partial capacity to  $u_c$ 
45:        Update  $\text{sat}_{u_c}(t)$ ; set remaining relay capacity to 0; break
46:      else
47:        break
48:      end if
49:    end for
50:  end for
51:  Validate single-association, beam power, satellite total power, and EPFD constraints
52:  Store  $\mathbf{x}^{\text{final}}(t)$ ,  $P_{s,b}^{\text{final}}(t)$ , and  $\text{sat}_u^{\text{final}}(t)$ 
53: end for

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G. Complexity Discussion

SAPR-R avoids directly solving the mixed-integer nonlinear optimization problem formulated in Chapter III. Since the beam relation record $\mathcal{R}_B(t)$ is precomputed, the online procedure does not need to re-identify the critical satellite, helper satellites, or beam-pair mappings in each time slot. The online computation is therefore limited to candidate construction, local sorting, power allocation, and sequential reassociation within predefined beam pairs.

Let K_t^{SAHR} denote the number of candidate helper users considered in SAHR at time slot t , and let K_t^{BPLR} denote the number of candidate closed-beam users considered in BPLR. In the SAHR stage, candidate construction and released-power computation are linear in K_t^{SAHR} , while sorting the candidates by released power dominates the complexity:

$$\mathcal{O}(K_t^{\text{SAHR}} \log K_t^{\text{SAHR}}).$$

In the relay-beam boosting stage, let H_t denote the number of helper relay beams in time slot t . Sorting helper relay beams by relay demand requires

$$\mathcal{O}(H_t \log H_t).$$

In the BPLR stage, candidate users are sorted by achievable relay service rate, and then allocated sequentially according to the remaining relay capacity. The dominant complexity is therefore

$$\mathcal{O}(K_t^{\text{BPLR}} \log K_t^{\text{BPLR}}).$$

Over all time slots, the online complexity is

$$\mathcal{O}\left(\sum_{t=1}^T (K_t^{\text{SAHR}} \log K_t^{\text{SAHR}} + H_t \log H_t + K_t^{\text{BPLR}} \log K_t^{\text{BPLR}})\right).$$

Thus, SAPR-R reduces the online decision process to local sorting and sequential allocation within precomputed beam relations, instead of solving a global association and continuous power-control problem in every time slot.

H. Summary

This chapter presented SAPR-R for maintaining user service under the GS EPFD constraint during critical time slots. The method first precomputes time-slot beam relation records from predictable satellite geometry, beam footprints, and beam-level EPFD information. During online operation, SAPR-R applies SAHR, helper relay beam boosting, and BPLR in sequence.

The staged design separates the original coupled problem into local decisions. SAHR uses critical safe beams to take over selected helper users and release helper satellite power. Helper relay beam boosting allocates the available relay power budget according to relay demand. BPLR then reassociates closed-beam users to feasible helper relay beams within predefined beam pairs. In this way, SAPR-R improves user satisfaction during critical periods while satisfying the EPFD constraint.

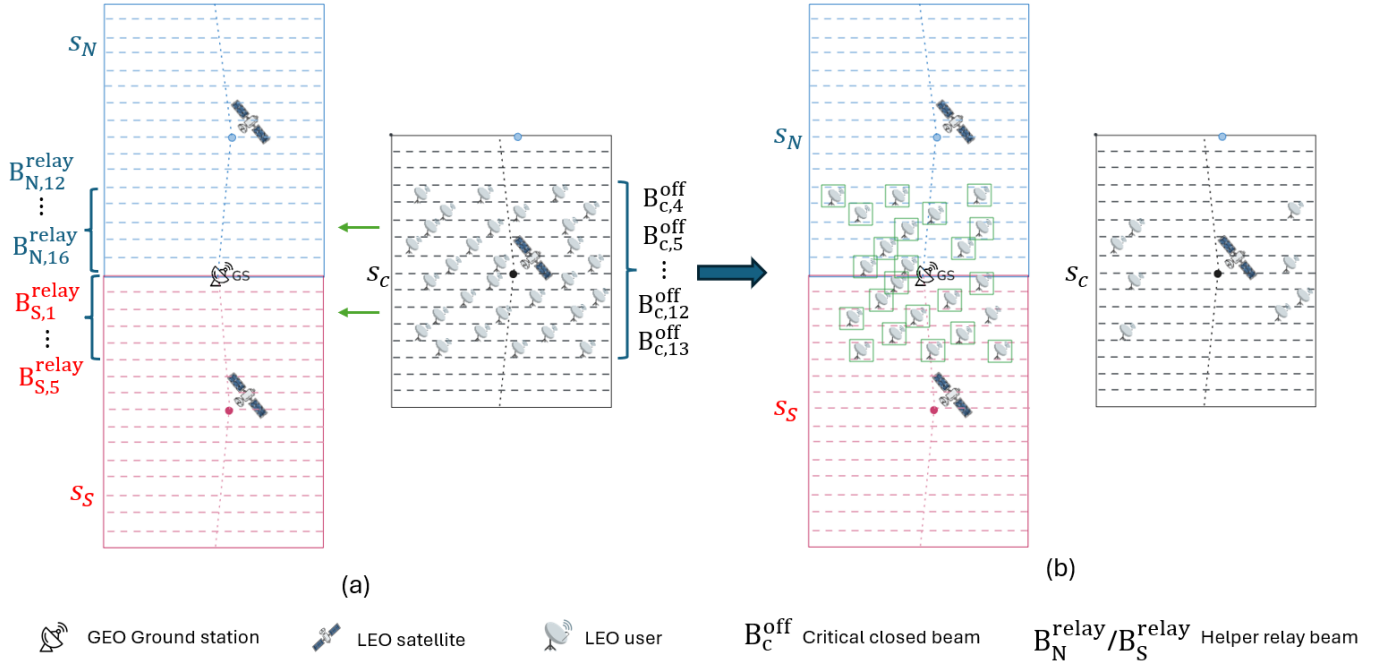


Fig. 4. Illustration of Beam-Pair Local Relay (BPLR). Candidate critical users in closed beams are reassociated to the corresponding helper relay beams according to $\mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)$.

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